A less well-known result is the following.

Theorem 5.5.1 *If $x > 0$ is a normal in base 2 and p, q are odd coprime positive integers, then px, qx are also normal in base 2. The correlation between the binary digits of px and qx is equal to*

$$\rho\Big(px, qx\Big) = \frac{1}{pq} = \rho\Big(x, \frac{q}{p}x\Big) = \rho\Big(x, \frac{p}{q}x\Big) \tag{5.29}$$

Proof
As a starting point, it is easy to show that for two normal numbers x, y, the correlation between their binary digit sequences $(d_n(x))$ and $(d_n(y))$ is given by the formula

$$\rho(x, y) = -1 + \lim_{n\to\infty} \frac{4}{n} \sum_{k=0}^{n-1} d_k(x)d_k(y). \tag{5.30}$$

Also, a normal number multiplied by a non-zero rational is normal in the same base (Wall's theorem, 1949; see also [9]). Thus, the binary digit distributions of x, px and qx have the same mean $\frac{1}{2}$ and same variance $\frac{1}{4}$. Proving (5.29) is more difficult. A proof was first published on CrossValidated.com in 2019, see here and here. The proof assumes that the binary digits of x are randomly distributed as an infinite Bernoulli trial with 50% of 0 and 1, a stronger assumption than normality in base 2 ∎

Now, if x is a normal number, $\alpha \neq \beta$ are positive integers and p, q are odd coprime positive integers, then we have the following (the proof is left as an exercise):

$$\rho\Big(2^\alpha px, 2^\beta qx\Big) = 0. \tag{5.31}$$

I now can state and prove the following deep result with important implications, for instance the fact that the binary digit sequences of $\sqrt{2}$ and $\sqrt{3}$ are uncorrelated, that is $\rho(\sqrt{2}, \sqrt{3}) = 0$, if both are normal numbers.

Theorem 5.5.2 *Let x, y be normal numbers in base 2, linearly independent over $\mathbb{Q}$. Then $\rho(x, y) = 0$.*

Proof
Linear independence over $\mathbb{Q}$ means that if $\alpha x = \beta y$ for some integers α, β, we must have $\alpha = \beta = 0$. There are infinitely many pairs of sequences $(\alpha_t), (\beta_t)$ such that $y_t = \alpha_t x/\beta_t \to y$ as $t \to \infty$, with α_t, β_t being positive odd coprimes for all t. By virtue of theorem 5.5.1, we have

$$\rho(x, y_t) = \frac{1}{\alpha_t \beta_t}.$$

As t increases, the number of identical digits on the left in x and y_t increases, and thus $\rho(x, y_t) \to \rho(x, y)$. At the same time, $\alpha_t, \beta_t \to \infty$ because there is no rational number r such that $x = ry$ due to x, y being linearly independent over $\mathbb{Q}$. Thus, $\rho(x, y) = 0$. ∎

Formulas (5.29) and (5.31) lead to a new test of randomness. For instance, to check if the binary digits of a number x are random enough, you first compute $\lambda_n(p, q) = \rho_n(px, qx)$, the empirical correlation on the first n digits, for various values of n, p, q. Then run N simulations, replacing the digits of x by N sequences of random bits. Now let $L_n(p, q)$ and $U_n(p, q)$ be the lower and upper correlations computed over the N samples with fixed p, q. If $\lambda_n(p, q) \notin [L_n(p, q), U_n(p, q]$ far more often than expected by chance, then the digits of x are presumed non-random. Also, if for some p, q, the value $(pq)^{-1}$ is not within these two bounds, it means that your random number generator is defective.

Theorem 5.5.2 is particularly useful in the context of strong PRNGs (pseudo-random number generators), with applications to cryptography where replicability is mandatory, or to test the strength of other PRNGs. In particular, the PRNG discussed in section 4.4 in [14] relies on a large number of quadratic irrationals: the square roots of square-free integers. Random bits are generated by

- choosing (say) 10^6 such numbers,
- for each of them extracting 10^4 binary digits starting at a random location in the digit expansion,
- then concatenate all the collected digits to produce 10^{10} random bits.

This PRNG offers up to 10^6 distinct seed pairs. Each pair consists of (1) a square-free integer and (2) the location or index where the binary digit sequence must start in the associated quadratic irrational. Then theorem 5.5.2 guarantees that the 10^6 digit sequences are uncorrelated, if the underlying square root numbers are normal.

Now I share my Python code to compute the correlation between the binary digits of px and qx, where x is a real number in $[0, 1]$ and p, q are positive integers, coprime or not, odd or even. The function `vg_correl`

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